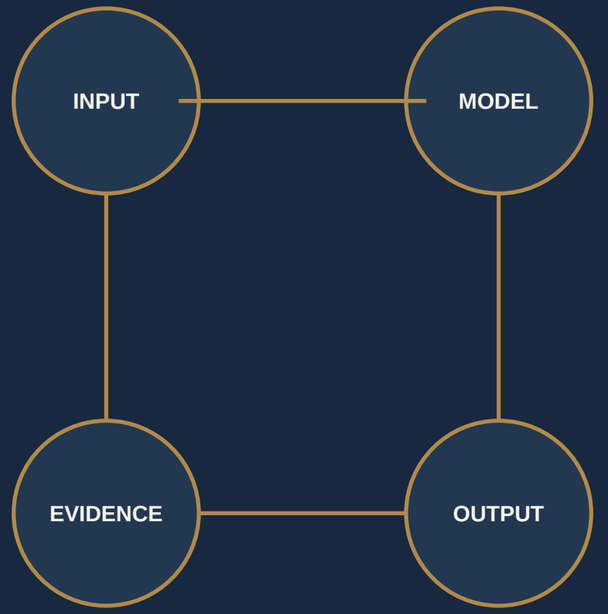
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Robot Kinematics

AI.212 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define coordinate transformations.
- Apply key methods including forward kinematics.
- Build or analyze a practical system involving inverse kinematics.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Coordinate transformations

02

METHODS

Forward kinematics

03

SYSTEMS

Inverse kinematics

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Coordinate transformations

Coordinate transformations

02 · Forward kinematics

Forward kinematics

03 · Inverse kinematics

Inverse kinematics

04 · Denavit-Hartenberg representat

Denavit-Hartenberg representation

05 · Robot configurations and wor

Robot configurations and workspace analysis

06 · Mandatory practical: Derive a

Mandatory practical: Derive and implement robot kinematic models and control simulated robot arms

07 · ### Year 3 — Semester 1

Year 3 — Semester 1

Core ideas, made explicit

AI.212 • Distinguish definitions, assumptions and evidence

- Coordinate transformations
- Forward kinematics
- Inverse kinematics
- Denavit-Hartenberg representation
- Robot configurations and workspace analysis

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.212 • Distinguish definitions, assumptions and evidence

- Mandatory practical: Derive and implement robot kinematic models and control simulated robot arms
- ### Year 3 — Semester 1

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
Coordinate transformations

INPUTS
Forward kinematics

METHOD
Inverse kinematics

OUTPUT
Denavit-Hartenberg representation

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving coordinate transformations.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using coordinate transformations.
- Implement or simulate a baseline using forward kinematics.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.